Clinical Readmission Pipeline - Executive Summary

Steps 1-3: EDA -> Cleaning -> Feature Engineering | Target: Readmission_30d | 2026-09-01

Step 1 - Data Collection & Exploration

Records loaded:

1,155,000 patients | 41 raw features

Target balance:

Not Readmitted: 84.98% (910,243) **Readmitted: 15.02% (160,918) - imbalanced**

Key issues found:

Duplicate case-mismatched columns (Age/age, Gender/gender, Drug_Name/drug_name); 100% null columns (Extra_Col_1/2); invalid values (Age up to 999, negative weights)

Outputs generated:

step1_clinical_distributions.png, step1_correlation_matrix.png

Step 2 - Data Cleaning & Preprocessing

- Column reduction: dropped 11 columns (nulls, duplicates, redundant) -> 41 to 30 columns
- Row validation: removed 37,988 rows with invalid Age/Weight
- Missing target rows: dropped 81,110 rows without a Readmission_30d label
- Imputation: numeric columns filled with median, categorical with mode
- Duplicates removed: 63,351 duplicate rows dropped
- Outlier capping (IQR): Age, BMI, Dosage, Hemoglobin, Creatinine, ALT/AST enzymes
- Encoding: label-encoded Gender; one-hot encoded Drug_Name and Route

Final output: data_cleaned.csv -> 972,551 rows x 553 columns

Data quality note: Gender retains messy raw values (MAL, FEMAL, O, N/A, etc.); high-cardinality Drug_Name one-hot encoding inflated column count. Recommend consolidating before modeling.

Step 3 - Feature Engineering

kidney_stage	eGFR-based staging (Normal/Mild/Moderate/Severe)
liver_risk	Flag if ALT or AST > 40
polypharmacy	Flag if 5+ concurrent medications
bmi_category	Underweight/Normal/Overweight/Obese
age_group	Pediatric through Elderly bands
elderly_high_dose	Age > 65 and above-median dosage
de_ritis_ratio	AST/ALT ratio (liver disease indicator)

Final output: data_engineered.csv -> 972,551 rows x 560 columns

Pipeline Status

Step	Status	Runtime Notes
1. EDA	Complete	No errors
2. Cleaning	Complete	Minor deprecation warnings only
3. Feature Engineering	Complete	Performance warnings (fragmentation) only, no data loss

Next Step: Step 4 - Model Training & Evaluation on data_engineered.csv